Design Guide for Saudi Arabia

Minimum design standards for mechanical services



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Design Guide for the Kingdom of Saudi Arabia:

Minimum design standards for mechanical services

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The Chartered Institution of Building Services Engineers
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Note from the publisher

This publication is primarily intended to provide guidance to those responsible for the design, installation, commissioning, operation and maintenance of building services. It is not intended to be exhaustive or definitive and it will be necessary for users of guidance to exercise their own professional judgement when deciding whether to abide by or depart from it.

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This document aims at establishing the minimum design standards for the design of mechanical systems for buildings in the Kingdom of Saudi Arabia. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

This guide has been produced in good faith by the CIBSE UAE Technical Committee to promote best engineering practices in the Kingdom of Saudi Arabia, without legal or financial liability of CIBSE UAE or the authors, in order to assist engineers coming from elsewhere in the world to work in the Kingdom of Saudi Arabia. It should be read alongside existing CIBSE guidance relevant to the region, including *Buildings for Extreme Environments: Arid*.

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1 Design codes, standards and guidance

The design, specification and installation of mechanical services should be in accordance with good engineering practice and should conform to local codes, standards and guidelines, and follow applicable international codes, standards and guidelines where indicated.

1.1 Local codes, standards and guidance

Table 1.1 Local codes, standards and guidance

Source/reference	Code, regulation, guideline reference
Saudi Building Code National Committee	SBC-201: *Saudi Building Code: General* (SBCNC, 2018a) SBC-501: *Saudi Mechanical Code* (SBCNC, 2018b) SBC-601: *Saudi Energy Conservation Code: Buildings Except Low-Rise (Residential) Buildings* (SBCNC, 2018c) SBC-602: *Saudi Energy Conservation Code Low Rise (Residential) Buildings* (SBCNC, 2018d)
Mostadam	Comprehensive sustainability rating and certification system for KSA (https://www.mostadam.sa)
Saudi Standards, Metrology and Quality Organization (SASO)	SASO 2663:2018: *Energy Labelling and Minimum. Energy Performance Requirements for Air-Conditioners* (SASO, 2018) SASO 2874:2016: *Large capacity air conditioners — performance requirements and methods of testing* (SASO, 2016)

1.2 International codes, standards and guidance

Table 1.2 International codes, standards and guidance

Source/reference	Code, regulation, guideline reference
American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)	ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017a) ASHRAE Standard 15: *Safety Standard for Refrigeration Systems* (ASHRAE, 2019a) ASHRAE Standard 55: *Thermal Environmental Conditions for Human Occupancy* (ASHRAE, 2020a) ASHRAE Standard 62.1: *Ventilation for Acceptable Indoor Air Quality* (ASHRAE, 2019b) ASHRAE Standard 62.2: *Ventilation for Acceptable Indoor Air Quality in Residential Buildings* ASHRAE, 2019c) ASHRAE Standard 90.1: *Energy Standard for Buildings Except Low-Rise Residential Buildings* (ASHRAE, 2019d) ASHRAE Standard 170: *Ventilation of Health Care Facilities* (ASHRAE, 2017b) ASHRAE 183: *Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings* (ASHRAE, 2020b)
Chartered Institution of Building Services Engineers (CIBSE)	CIBSE Commissioning Code A: *Air distribution systems* (2006) CIBSE Commissioning Code W: *Water distribution systems* (2010) CIBSE Commissioning Code L: *Lighting* (2018) CIBSE Commissioning Code C: *Automatic controls* (2001) CIBSE Commissioning Code R: *Refrigeration systems* (2002a) CIBSE Commissioning Code B: *Boilers* (2002b)

Table continues

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Table 1.2 International codes, standards and guidance (*continued*)

Source/reference	Code, regulation, guideline reference
Chartered Institution of Building Services Engineers (CIBSE) (*continued*)	CIBSE Guide A: *Environmental design* (CIBSE, 2015) CIBSE Guide B4: *Noise and vibration control for building services systems* (CIBSE, 2016)
Sheet Metal Air Conditioning Contractors National Association (SMACNA)	Various guidance publications (https://www.smacna.org)
Building Engineering Services Association (BESA)	DW144: *Specification for sheet metal ductwork* (BESA, 2016) DW172: *Specification for kitchen ventilation systems* (BESA, 2018)
British Standards Institution (BSI)	BS 5422: *Method for specifying thermal insulating materials for pipes, tanks, vessels, ductwork and equipment operating within the temperature range –40 °C to +700 °C* (BSI, 2009) BS 8233: *Guidance on sound insulation and noise reduction for buildings* (BSI, 2014)
Education Funding Agency (UK)	Building Bulletin 93: *Acoustic design of schools — performance standards* (EFA, 2015)

2 KSA climate zones

The kingdom of Saudi Arabia is considered to be made up of three climate zones as represented by Figure 2.1. Selected cities are listed by zone in the Table 2.1.

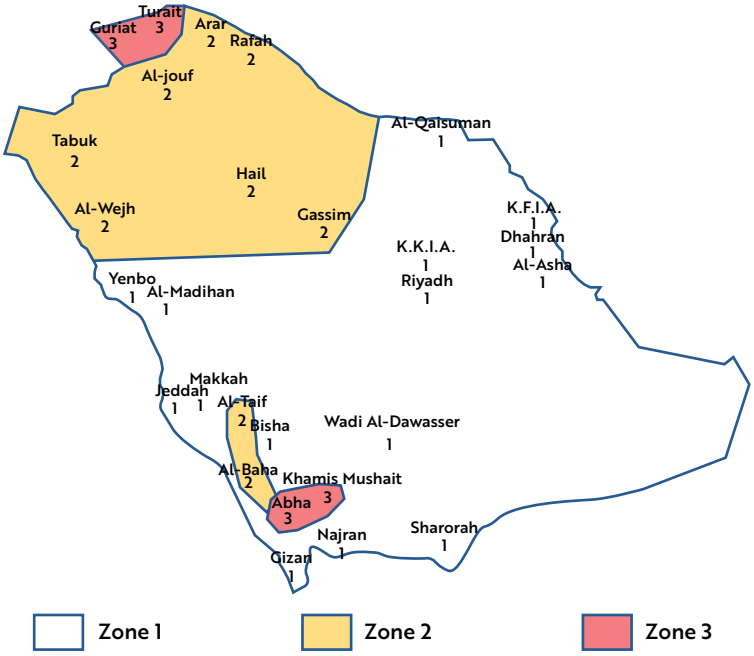


Figure 2.1 KSA climate zones (source: *Saudi Energy Conservation Code: Buildings Except Low-Rise (Residential) Buildings*, chapter 11 (SBCNC, 2018c))

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Table 2.3 KSA climate zones (source: *Saudi Energy Conservation Code: Buildings Except Low-Rise (Residential) Buildings*, chapter 11 (SBCNC, 2018c))

Location			Zone according to SBC 601/602	Zone according to IECC	Elevation (m)	Dry bulb temp. (10-year average) (°C)		CDD10
Name	Latitude	Longitude				1%	Max.	
Abha	18.23N	42.65E	3	3A and 3B	2093	30.3	34	3289
Al-Ahsa	25.30N	49.48E	1	1	178	46	49.8	6389
Al-Baha	20.30N	41.65E	2	2	1652	35.2	39	4781
Al-Jouf	29.78N	40.10E	2	2	689	41.8	46.7	4612
Al-Madinah	24.55N	39.70E	1	1	636	44.2	47.7	6715
Al-Qaisumah	28.32N	46.13E	1	1	358	45.2	50.4	5706
Al-Taif	21.48N	40.55E	2	2	1453	36.1	41	4804
Al-Wejh	26.20N	36.48E	1	1	24	34.2	44.7	5493
Arar	30.90N	41.13E	2	2	549	42.3	47.3	4571
Bisha	19.98N	42.63E	1	1	1162	39.8	42.8	5640
Dhahran	26.27N	50.17E	1	1	17	44	49.3	6258
Gassim	26.30N	43.77E	1	1	648	44	48.1	5545
Gizan	16.88N	42.58E	1	1	7	37.9	42.3	7451
Guriat	31.40N	37.28E	2	2	504	39.1	45.4	3729
Hail	27.43N	41.68E	2	2	1002	40.9	45.2	4547
Jeddah (King Abdul Aziz Int.	21.70N	39.18E	1	1	17	39.8	48.3	6803
Jubail	27.00N	49.67E	1	1	5	40.0	49.3*	6258*
King Faha Int. Airport	26.45N	49.82E	1	1	12	44.6	49.2	6274
Khamis Mushait	18.30N	42.80E	2	2	2056	31.1	36.9	3710
King Khaled Int. Airport	24.93N	46.72E	1	1	614	43.9	47.6	5756
Makkah	21.43N	39.77E	1	1	240	44.1	49.2	7800
Najran	17.62N	44.42E	1	1	1212	39.2	42.7	5643
Rafha	29.62N	43.48E	2	2	444	43.7	48.2	5000
Riyadh Obs. (O.A.P.)	24.70N	46.73E	1	1	620	43.9	47.2	6107
Sharorah	17.47N	47.10E	1	1	725	42.2	45.1	6579
Tabuk	28.38N	36.60E	2	2	768	39.8	44.2	4508
Turaif	31.68N	38.73E	2	2	852	38.8	43.8	3542
Wadi Al-Dawasser Al	20.50N	45.25E	1	1	622	44	47.8	6591
Yenbo	24.13N	38.07E	1	1	10	42.2	48.9	6626

* Dhahran data used

As presented in the *Saudi Energy Conservation Code* (SBCNC, 2018c), the above table provides temperatures based on ASHRAE weather data for 1% annual cooling condition. However, this *Design Guide* encourages the designer to use values corresponding to 0.4% in the event of optimizing thermal comfort during high temperature hours.

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3 Minimum building fabric parameters

The minimum building fabric parameters are governed by the requirements stated in Saudi Building Codes SBC-601 (buildings except low-rise) (SBCNC, 2018c) and SBC-602 (low rise residential buildings) (SBCNC, 2018d). The various envelope components and space types are represented in Figure 3.1 below.

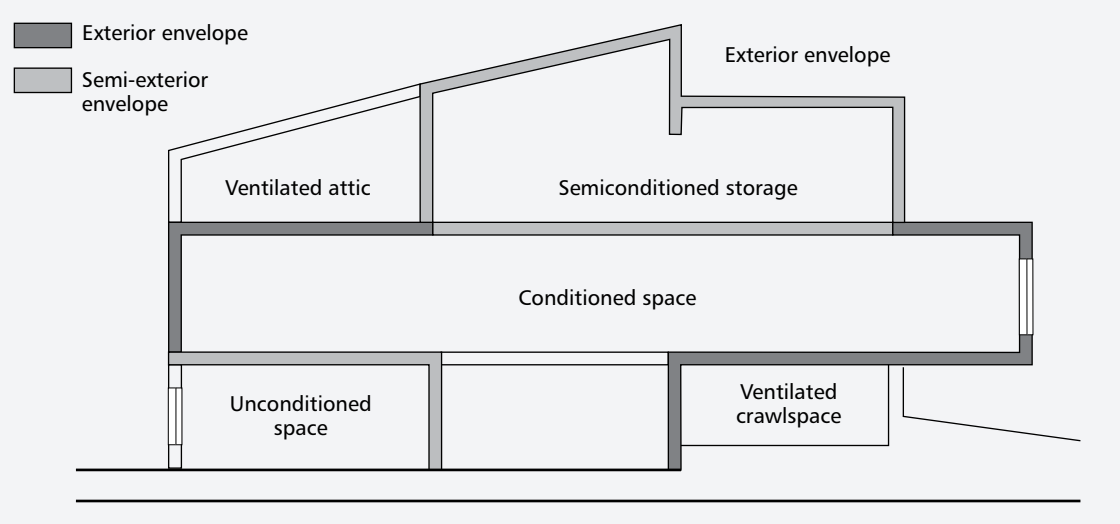


Figure 3.1 Envelope and space types

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3.1 Buildings other than low-rise residential

Tables 3.1 to 3.3 below represent minimum fabric parameters in accordance to SBC-601 (SBCNC, 2018c) and based on the climate zone classification.

Table 3.1a Climate Zone 1: Building envelope requirements — opaque elements

Ref.	Opaque elements	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs						
(a)	Insulation entirely above deck	0.357	2.66 (CI)	0.272	3.54 (CI)	1.250	0.67 (CI)
(b)	Metal buildings	0.369	2.57 (CI)	0.369	2.57 (CI)	0.950	0.92 (CI)
(c)	Attic with steel joists	0.193	6.91	0.153	9.70	0.460	2.27
2	Walls, above-grade						
(a)	Mass	1.191	0.60 (CI)	0.312	2.96 (CI)	1.191	0.60 (CI)
(b)	Metal buildings	0.193	5.04 (CI)	0.193	5.04 (CI)	0.233	4.14 (CI)
(c)	Steel framed	0.255	3.42 (CI)	0.255	3.42 (CI)	0.738	0.86 (CI)
3	Walls, below-grade						
(a)	Wall	1.305†	0.62 (CI)	1.305†	0.62 (CI)	1.305†	0.62 (CI)
4	Floors						
(a)	Mass	0.397	1.97 (CI)	0.397	1.97 (CI)	0.397	1.97 (CI)
(b)	Steel joist	0.408	1.94 (CI)	0.408	1.94 (CI)	0.408	1.94 (CI)
5	Floor slab, on-grade						
(a)	Unconditioned	0.727*	1.18 (FI)	0.727*	1.18 (FI)	0.727*	1.18 (FI)
6	Opaque doors						
(a)	Swinging	2.84	—	2.84	—	2.84	—
(b)	Non-swinging	5.90	—	5.90	—	5.90	—

* *F*-factor: floor conductance per linear metre (W/m·K)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated

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Table 3.1b Climate Zone 1: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value (W/m²·K)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly max. solar heat gain coeff.
1	Vertical glazing (0% to 40% of wall)						
(a)	Non-metal framing (all)	2.89	0.25	2.89	0.25	2.89	NR
(b)	Metal framing (curtain wall/store front)	2.89		2.89		2.89	
(c)	Metal framing (curtain store front)	2.89		2.89		2.89	
(d)	Metal framing (all other)	2.89		2.89		2.89	
2	Vertical glazing (>40% to 50% of wall)						
(a)	Non-metal framing (all)	2.38	0.25	2.38	0.25	2.38	NR
(b)	Metal framing (curtain wall/store front)	2.38		2.38		2.38	
(c)	Metal framing (curtain store front)	2.38		2.38		2.38	
(d)	Metal framing (all other)	2.38		2.38		2.38	
3	Vertical glazing (>50% to 100% of wall)						
(a)	Non-metal framing (all)	1.89	0.25	1.89	0.25	1.89	NR
(b)	Metal framing (curtain wall/store front)	1.89		1.89		1.89	
(c)	Metal framing (curtain store front)	1.89		1.89		1.89	
(d)	Metal framing (all other)	1.89		1.89		1.89	
4	Skylight with curb, glass						
(a)	0%–2.0% of roof	11.23	0.36	11.23	0.19	11.23	NR
(b)	2.1%–5.0% of roof	11.23	0.19	11.23	0.16	11.23	
5	Skylight with curb, plastic						
(a)	0%–2.0% of roof	10.78	0.34	10.78	0.27	10.78	NR(
(b)	2.1%–5.0% of roof	10.78	0.27	10.78	0.27	10.78	
6	Skylight without curb, glass						
(a)	0%–2.0% of roof	7.72	0.36	7.72	0.19	7.72	NR
(b)	2.1%–5.0% of roof	7.72	0.19	7.72	0.19	7.72	

NR = not required

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Table 3.2a Climate Zone 2: Building envelope requirements — opaque elements

Ref.	Opaque elements	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs						
(a)	Insulation entirely above deck	0.454	2.06 (CI)	0.340	2.80 (CI)	1.590	0.49 (CI)
(b)	Metal buildings	0.471	1.99 (CI)	0.470	1.99 (CI)	1.135	0.74 (CI)
(c)	Attic with steel joists	0.244	5.00	0.190	6.91	0.579	1.73
2	Walls, above-grade						
(a)	Mass	1.362	0.49 (CI)	0.352	2.616 (CI)	1.362	0.49 (CI)
(b)	Metal buildings	0.216	4.49 (CI)	0.216	4.49 (CI)	0.267	3.59 (CI)
(c)	Steel framed	0.284	3.03 (CI)	0.284	3.03 (CI)	0.851	0.67 (CI)
3	Walls, below-grade						
(a)	Wall	1.305 †	0.62 (CI)	1.305 †	0.62 (CI)	1.305 †	0.62 (CI)
4	Floors						
(a)	Mass	0.567	1.22 (CI)	0.567	1.22 (CI)	0.567	1.22 (CI)
(b)	Steel joist	0.624	1.09 (CI)	0.624	1.09 (CI)	0.624	1.09 (CI)
5	Floor slab, on-grade						
(a)	Unconditioned	0.727*	1.18 (FI)	0.727*	1.18 (FI)	0.727*	1.18 (FI)
6	Opaque doors						
(a)	Swinging	2.84	—	2.84	—	2.84	—
(b)	Non-swinging	5.90	—	5.90	—	5.90	—

* *F*-factor: floor conductance per linear metre (W/m·K)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated

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Table 3.2b Climate Zone 2: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value (W/m²·K)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly max. solar heat gain coeff.
1	Vertical glazing (0% to 40% of wall)						
(a)	Non-metal framing (all)	2.89	0.25	2.89	0.25	2.89	NR
(b)	Metal framing (curtain wall/store front)	2.89		2.89		2.89	
(c)	Metal framing (curtain store front)	2.89		2.89		2.89	
(d)	Metal framing (all other)	2.89		2.89		2.89	
2	Vertical glazing (>40% to 50% of wall)						
(a)	Non-metal framing (all)	2.38	0.25	2.38	0.25	2.38	NR
(b)	Metal framing (curtain wall/store front)	2.38		2.38		2.38	
(c)	Metal framing (curtain store front)	2.38		2.38		2.38	
(d)	Metal framing (all other)	2.38		2.38		2.38	
3	Vertical glazing (>50% to 100% of wall)						
(a)	Non-metal framing (all)	1.87	0.25	1.87	0.25	1.87	NR
(b)	Metal framing (curtain wall/store front)	1.87		1.87		1.87	
(c)	Metal framing (curtain store front)	1.87		1.87		1.87	
(d)	Metal framing (all other)	1.87		1.87		1.87	
4	Skylight with curb, glass						
(a)	0%–2.0% of roof	11.23	0.36	11.23	0.19	11.23	NR
(b)	2.1%–5.0% of roof	11.23	0.19	11.23	0.16	11.23	
5	Skylight with curb, plastic						
(a)	0%–2.0% of roof	10.78	0.34	10.78	0.27	10.78	NR
(b)	2.1%–5.0% of roof	10.78	0.27	10.78	0.27	10.78	
6	Skylight without curb, glass						
(a)	0%–2.0% of roof	7.72	0.36	7.72	0.19	7.72	NR
(b)	2.1%–5.0% of roof	7.72	0.19	7.72	0.19	7.72	

NR = not required

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Table 3.3a Climate Zone 3: Building envelope requirements — opaque elements

Ref.	Opaque elements	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs						
(a)	Insulation entirely above deck	0.511	1.81 (CI)	0.397	2.38 (CI)	1.759	0.42 (CI)
(b)	Metal buildings	0.528	1.76 (CI)	0.528	1.76 (CI)	1.362	0.60 (CI)
(c)	Attic with steel joists	0.272	4.40	0.216	5.96	0.647	1.50
2	Walls, above-grade						
(a)	Mass	1.588	0.39 (CI)	0.414	2.17 (CI)	1.588	0.39 (CI)
(b)	Metal buildings	0.255	3.77 (CI)	0.255	3.77 (CI)	0.312	3.05 (CI)
(c)	Steel framed	0.34	2.43 (CI)	0.340	2.43 (CI)	0.964	0.53 (CI)
3	Walls, below-grade						
(a)	Wall	1.305†	0.62 (CI)	1.305†	0.62	1.305†	0.62 (CI)
4	Floors						
(a)	Mass	0.567	1.22 (CI)	0.567	1.22 (CI)	0.567	1.22 (CI)
(b)	Steel joist	0.624	1.09 (CI)	0.624	1.09 (CI)	0.624	1.09 (CI)
5	Floor slab, on-grade						
(a)	Unconditioned	0.727*	1.18 (FI)	0.727*	1.18 (FI)	0.727*	1.18 (FI)
6	Opaque doors						
(a)	Swinging	2.84	—	2.84	—	2.84	—
(b)	Non-swinging	5.90	—	5.90	—	5.90	—

* *F*-factor: floor conductance per linear metre (W/m·K)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated

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Table 3.3b Climate Zone 3: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential		Semi-conditioned	
		Assembly max. *U*-value (W/m²·K)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly solar heat gain coeff.	Assembly max. *U*-value (W/m²·K)	Assembly max.solar heat gain coeff.
1	Vertical glazing (0% to 40% of wall)						
(a)	Non-metal framing (all)	2.89	0.25	2.89	0.25	2.89	NR
(b)	Metal framing (curtain wall/store front)	2.89		2.89		2.89	
(c)	Metal framing (curtain store front)	2.89		2.89		2.89	
(d)	Metal framing (all other)	2.89		2.89		2.89	
2	Vertical glazing (>40% to 50% of wall)						
(a)	Non-metal framing (all)	2.38	0.25	2.38	0.25	2.38	NR
(b)	Metal framing (curtain wall/store front)	2.38		2.38		2.38	
(c)	Metal framing (curtain store front)	2.38		2.38		2.38	
(d)	Metal framing (all other)	2.38		2.38		2.38	
3	Vertical glazing (>50% to 100% of wall)						
(a)	Non-metal framing (all)	1.87	0.25	1.87	0.25	1.87	NR
(b)	Metal framing (curtain wall/store front)	1.87		1.87		1.87	
(c)	Metal framing (curtain store front)	1.87		1.87		1.87	
(d)	Metal framing (all other)	1.87		1.87		1.87	
4	Skylight with curb, glass						
(a)	0%–2.0% of roof	11.23	0.36	11.23	0.19	11.23	NR
(b)	2.1%–5.0% of roof	11.23	0.19	11.23	0.16	11.23	
5	Skylight with curb, plastic						
(a)	0%–2.0% of roof	10.78	0.34	10.78	0.27	10.78	NR
(b)	2.1%–5.0% of roof	10.78	0.27	10.78	0.27	10.78	
6	Skylight without curb, glass						
(a)	0%–2.0% of roof	7.72	0.36	7.72	0.19	7.72	NR
(b)	2.1%–5.0% of roof	7.72	0.19	7.72	0.19	7.72	

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3.2 Low-rise buildings

Tables 3.4 to 3.6 below represent minimum fabric parameters in accordance to SBC-602 (SBCNC, 2018d) and based on the climate zone classification.

Table 3.4a Climate Zone 1: Building envelope requirements — opaque elements

Ref.	Opaque elements	Residential, conditioned		Residential, unconditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs				
(a)	Insulation entirely above deck	0.202	5.0 (CI)	0.4	2.5 (CI)
2	Walls, above-grade				
(a)	Mass (continuous insulation)	0.342	2.92 (CI)	0.453	2.20 (CI)
3	Walls, below-grade				
(a)	Wall	0.678	1.3 (CI)	6.473	NR
4	Floors				
(a)	Mass	0.496	1.50 (CI)	0.78	0.70 (CI)
(b)	Steel joist	0.296	3.30 (CI)	0.296	3.3 (CI)
(c)	Others	0.188	5.30 (CI)	0.288	3.30 (CI)
5	Floor slab, on-grade				
(a)	Slab (for 60 cm)	0.90*	2.6 (FI)	1.263*	NR
6	Opaque doors				
(a)	All assemblies	2.839	—	2.839	—

* *F*-factor: floor conductance per linear metre (W/m·K)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated; NR = not required

Table 3.4b Climate Zone 1: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.
1	Vertical glazing, 25% of wall				
(a)	All assemblies	2.668	0.25	3.695	0.25
2	Skylight with curb, glass				
(a)	All types, 0%–3.0% of roof	4.259	0.35	10.22	0.35

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Table 3.5a Climate Zone 2: Building envelope requirements — opaque elements

Ref.	Opaque elements	Residential, conditioned		Residential, unconditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs				
(a)	Insulation entirely above deck	0.238	4.2 (CI)	0.453	2.2 (CI)
2	Walls, above-grade				
(a)	Mass (continuous insulation)	0.397	2.5 (CI)	0.513	1.95 (CI)
3	Walls, below-grade				
(a)	Wall	0.678†	1.3 (CI)	6.473†	NR
4	Floors				
(a)	Mass	0.496	1.50 (CI)	0.780	0.70 (CI)
(b)	Steel joist	0.296	3.30 (CI)	0.296	3.30 (CI)
(c)	Others	0.188	5.30 (CI)	0.288	3.30 (CI)
5	Floor slab, on-grade				
(a)	Slab (for 60 cm)	0.90*	2.6 (FI)	1.263*	NR
6	Opaque doors				
(a)	All assemblies	2.839	—	2.839	—

* *F*-factor: floor conductance per linear metre ($W/m \cdot K$)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated; NR = not required

Table 3.5b Climate Zone 2: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.
1	Vertical glazing, 25% of wall				
(a)	All assemblies	2.668	0.25	3.695	NR
2	Skylight with curb, glass				
(a)	All types, 0%–3.0% of roof	4.259	0.35	10.22	0.35

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Table 3.6a Climate Zone 3: Building envelope requirements — opaque elements

Ref.	Opaque elements	Residential, conditioned		Residential, unconditioned	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)	Assembly max. *U*-value ($W/m^2 \cdot K$)	Insulation min. *R*-value ($m^2 \cdot K/W$)
1	Roofs				
(a)	Insulation entirely above deck	0.273	3.5 (CI)	0.527	1.8 (CI)
2	Walls, above-grade				
(a)	Mass (continuous insulation)	0.453	2.3 (CI)	0.592	1.70 (CI)
3	Walls, below-grade				
(a)	Wall	0.678†	1.3 (CI)	6.473†	NR
4	Floors				
(a)	Mass	0.496	1.50 (CI)	0.780	0.70 (CI)
(b)	Steel joist	0.296	3.30 (CI)	0.296	3.30 (CI)
(c)	Others	0.188	5.30 (CI)	0.288	3.30 (CI)
5	Floor slab, on-grade				
(a)	Slab (for 60 cm)	0.90*	2.6 (FI)	1.263*	NR
6	Opaque doors				
(a)	All assemblies	2.839	—	2.839	—

* *F*-factor: floor conductance per linear metre ($W/m \cdot K$)

† *C*-factor: thermal conductance ($W/m^2 \cdot K$)

CI = continuous insulation; FI = fully insulated; NR = not required

Table 3.6b Climate Zone 3: Building envelope requirements — fenestration

Ref.	Fenestration	Non-residential		Residential	
		Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.	Assembly max. *U*-value ($W/m^2 \cdot K$)	Assembly max. solar heat gain coeff.
1	Vertical glazing, 25% of wall				
(a)	All assemblies	2.668	0.25	3.695	NR
2	Skylight with curb, glass				
(a)	All types, 0%–3.0% of roof	4.259	0.35	10.22	0.35

NR = not required

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4 External design temperatures

The external design conditions for the Kingdom of Saudi Arabia vary based on the climatic zone as indicated in preceding section. Table 4.1 represents the external condition of some of the major cities and at least one city in each climatic zone.

Table 4.1 External design conditions (source: ASHRAE *Fundamentals* (ASHRAE, 2017a))

Ref.	Reference city	Annual cooling (based on 0.4% annual occurrence)		Annual de-humidification (based on 0.4% annual occurrence)		Annual heating (based on 99.6% annual occurrence)		Annual humidification (based on 99.6% annual occurrence)	
		Dry bulb design temp. (°C)	Mean coincident wet bulb temp. (°C)	Dew point design temp. (°C)	Mean coincident dry bulb temp. (°C)	Dry bulb design temp. (°C)	Coldest month	Dew point design temp. (°C)	Mean coincident dry bulb temp. (°C)
1	Climate Zone 1								
(a)	Riyadh	44.7	19.3	17.9	22.9	5.9	January	–13.1	19.5
(b)	Jeddah	41.0	23.7	28.8	33.6	15.8	January	1.9	26.8
(c)	Dammam	46.0	20.9	27.3	32.8	6.9	January	–7.9	18.5
(d)	Makkah	45.2	24.4	26.8	35.3	16.3	January	3.0	26.4
(e)	Al Wajh	35.9	24.5	29.2	32.5	12.2	January	–1.2	18.7
2	Climate Zone 2								
(a)	Al Turaif	40.2	18.1	15.8	22.8	1.9	January	–11.9	17.3
(b)	Tabuk	41.1	18.9	15.2	27.3	1.9	January	–11.9	17.3
(c)	Al Jouf	43.2	19.0	14.1	21.7	1.0	January	–14.8	12.2
3	Climate Zone 3								
(a)	Abha	31.2	13.2	18.8	22.1	6.5	January	–12.0	15.3

Notes:
(1) Space conditioning for cooling should be sized based on values under 'annual cooling' design conditions.
(2) Space conditioning for heating should be sized based on values under 'annual heating' design conditions.
(3) Building cooling loads related to outdoor air treatment sized based on values under 'annual dehumidification'.
(4) Building heating loads related to outdoor air treatment sized based on values under 'annual humidification'.
(5) For equipment such as cooling towers and evaporative coolers, the sizing should be based on design conditions based on wet bulb temperatures. Refer to weather data.

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5 Cooling load calculations

The building cooling and heating load calculations for the purpose of sizing systems and equipment shall be determined in accordance with generally accepted engineering standards and handbooks acceptable to the respective local authority (for example ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017a).

6 Thermal loads

People density should be based on the architectural furniture arrangement, and where this is not available should then be based on SBC-601/602 (SBCNC, 2018c/d) or latest ASHRAE Standard 62.1. The associated sensible and latent loads should be based on ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017a).

Equipment heat gains shall be based on actual equipment heat dissipation of project specific components. In absence of that, ASHRAE Handbook: *Fundamentals* shall be used.

7 Lighting power density

7.1 Interior lighting

For new buildings except low-rise/residential, the average lighting power density for the interior connected lighting load must not exceed the values shown in Saudi Building Code SBC-601 Table 9.2, 'Using Building area method' or Table 9.3, 'Using space by space method' in Chapter 9, 'Lighting' (SBCNC, 2018c).

For residential buildings, the lighting power density should not exceed 10 watts/m^2 in accordance with Saudi Building Code SBC-602 Chapter 9, 'Lighting' (SBCNC, 2018d).

7.2 Exterior lighting

For all new buildings, the total exterior lighting power allowance for all exterior building applications should not exceed the values prescribed under Saudi Building Code SBC-601 Table 9.1B, Chapter 9 (SBCNC, 2018c).

8 Ventilation and infiltration

8.1 Infiltration rates

The design should take into account infiltration as specified in Saudi Building Code SBC-601 (SBCNC, 2018c), with a maximum leakage rate not exceeding the values shown in Table 8.1 below.

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Table 8.1 Air leakage rates as per Saudi Building Code SBC-601 (SBCNC, 2018c)

Ref.	Building fabric component	Differential pressure, ΔP (Pa)	Air leakage rate ($m^3 \cdot h^{-1}/m^2$)	Testing standard
1	Glazed swinging entrance doors and revolving doors	75	18.3	AAMA A440 (AAMA, 2008); NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
2	Curtain wall and storefront glazing	75	1.1	NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
3	Skylights	75	5.5	AAMA A440 (AAMA, 2008); NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
		300	9.1	AAMA A440 (AAMA, 2008); NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
4	Non-swinging opaque doors and glazed sectional garage doors	75	7.3	DASMA 105 (DASMA, 1992); NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
5	All other products	75	3.7	AAMA A440 (AAMA, 2008); NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
		300	5.5	

Air leakage shall be labelled and certified by the manufacturer.

For residential projects complying with Saudi Building Code SBC-602 (SBCNC, 2018c), the air leakage rates should not exceed the values shown in Table 8.2.

Table 8.2 Air leakage rates as per Saudi Building Code SBC-602 (SBCNC, 2018d)

Ref.	Building fabric component	Differential pressure, ΔP (Pa)	Air leakage rate ($m^3 \cdot h^{-1}/m^2$)	Testing standard
1	Fenestration and skylights	75	1.5	NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
2	Sliding doors	75	1.5	NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent
3	Swinging doors	75	2.5	NFRC 400 (NFRC, 2017); SASO E283 (SASO, 2012) or equivalent

As a minimum, an infiltration rate of 0.15 air changes per hour (ACH) is recommended to be used in the thermal modelling exercise in order to predict the building cooling loads. However, for airside equipment sizing, it is recommended to use 0.25 ACH for the areas near façades/external walls and 0.5 ACH for doors opening to the outside or higher depending on the door opening size.

8.2 Ventilation rates

The minimum outdoor airflow rate shall be determined in accordance with SBC-501 Section 403.

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9 Diversity and safety factor

The building calculations should also account for diversity profiles for people, lighting, equipment, etc., and should be considered in the air conditioning calculations. The local codes do not necessitate inclusion of a safety factor for cooling load calculations. However, sensible and latent safety factors of 10% and 5% respectively can be applied based on good engineering practice in calculating space by space heat load and sizing air conditioning equipment.

10 Building pressurisation

The air conditioning and ventilation systems should be designed such that the building will be maintained at a positive pressure. Generally the building should be pressurised with respect to the outside by approximately 10%; the outside air quantity should be 10% higher than the exhaust air quantity. This will assist in limiting infiltration of unconditioned ambient air.

11 Air movement

The air movement within the occupied zone should be designed to be in accordance with CIBSE recommendations or ASHRAE 55 (ASHRAE, 2020a) to prevent draughts and the associated feelings of discomfort and should be within the prescribed limits of 0.20 m/s to 0.30 m/s.

12 Noise and vibration control

Building services should be designed and installed as per chapter F08, 'Sound and vibration' of the ASHRAE *Fundamentals* Handbook (ASHRAE, 2017a), CIBSE Guide B4: *Noise and vibration control for building services systems* (CIBSE, 2016), CIBSE Guide A: *Environmental design* (CIBSE, 2015) and the advice of an acoustic consultant employed on the project. There are no specific room noise criteria provided in Saudi Building Codes SBC-601/602 (SBCNC, 2018c/d). Instead, SBC-601/602 calls for ventilation and air conditioning-related sound in rooms should be 3 to 35 dBA or room criterion (RC) 25 to 35.

13 Energy conservation

Systems should be designed to minimise energy using the guidance available in and SBC-601. There are no specific energy conservation methods prescribed in SBC-602 for the buildings which qualify as 'low-rise' residential buildings.

Energy conservation methods should be considered for all mechanical systems, in particular fans and pumps, which should be provided with variable speed drives for accurate commissioning and control where possible/ applicable to achieve energy saving targets.

For air handling units (AHUs) that comprise supply and extract in one unit, consideration should be given to energy recovery systems. Thermal wheel, plate heat exchangers and heat pipes should be considered for recovery of heat and the relative merits, including a quantitative assessment of efficiencies

Demand control ventilation is to be applied in all areas that have a design occupancy for ventilation of greater than 40 people per 100 m^2 and for spaces larger than 50 m^2. Occupancy density from architectural furniture arrangement should be considered, and where this is not available occupancy should be based on Table 403.3.1.1 of Saudi

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Mechanical Code SBC-501 (SBCNC, 2018b) or latest ASHRAE Standard 62.1 (ASHRAE, 2019b). Demand control ventilation can include the installation of carbon dioxide (CO_2) sensors, which should control and minimise the amount of outside air supplied to the space while maintaining the required CO_2 levels.

Effective energy management requires close and consistent control of all energy-consuming systems and components.

14 Cooling and refrigeration

14.1 Refrigerants

All refrigerants should be environmentally friendly with zero ozone depletion potential and low global warming potential (<100).

14.2 Cooling towers

Cooling towers should be selected to achieve the minimum energy performance requirements of the latest version of ASHRAE 90.1 (ASHRAE,2019d). All cooling towers should be CTI certified* and should be selected such that the minimum approach is at least 2.8 °C above the wet bulb temperature. Wet bulb temperatures for the specific city shall be based on ASHRAE weather data. Factory-built cooling towers shall be listed in accordance with UL Standard 1995: *Heating and Cooling Equipment* (UL, 2015) as per Saudi Mechanical Code SBC-501 (SBCNC, 2018b).

14.3 Chiller energy efficiency ratio (EER)

Chiller EER should be in accordance with SASO 2874-2016: *Large capacity air conditioners — performance requirement and methods of testing* (SASO, 2016).

15 Hydraulic pressures and temperature

Water pipework and ductwork system operating/working and test pressures and temperatures should be determined in accordance with the design parameters set by the mode of operation of that particular system and in full compliance with the appropriate specification, British Standard, code of practice or CIBSE guides, SMACNA guidance and all relevant standards and regulations.

16 Fire and life safety systems

The ventilation systems associated with fire and life safety (FLS) should be informed by the FLS strategy for the building.

Smoke venting provisions required for the building are determined based on Saudi General Building Code SBC-201 (SBCNC, 2018a) or Saudi Fire Protection Code SBC-801 (SBCNC, 2018e). The smoke control requirements detailed under SBC-501 document highlights the minimum requirements for the design, installation and acceptance testing of smoke control systems.

The smoke control systems are generally classified as below:

— smoke containment systems achieved by means of pressure differences across smoke

* Cooling Technology Institute (https://www.coolingtechnology.org/cti-certification)

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barriers such as stair pressurization, hoistway pressurization etc.

— smoke exhaust systems in compliance with NFPA 92

— other smoke control systems such as enclosed garage areas.

As per SBC-501, for enclosed parking garages, the ventilation systems shall be capable of producing a ventilation airflow rate 3.8 litre·s^{-1}/m^2. The systems shall be automatic and operating based on carbon monoxide detectors in conjunction with nitrogen dioxide detectors.

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